

- ① A photon emitted during the de-excitation of an electron from the state n to the first excited state in a hydrogen atom, irradiates a metallic cathode of work function 2 eV , in a photo cell, with a stopping potential of 0.55 V . Obtain the value of the quantum number of the state n .
- ② The magnetic susceptibility of magnesium at 300 K is 1.2×10^5 . At what temperature will it become 1.44×10^5 ?
- ③ Draw equipotential surfaces for an electric dipole?
- ④ Two wavelengths of sodium light of 590 nm and 596 nm are used in turn to study the diffraction taking place at a single slit of aperture $2 \times 10^{-6}\text{ m}$. If the distance between the slit and the screen is 1.5 m . Calculate the separation between the positions of second maxima of diffraction pattern observed in the two cases.
- ⑤ A $100\text{ }\mu\text{F}$ capacitor having plate separation of 4 mm is charged by 200 V dc. The source is now disconnected. When the distance between the plates is doubled and a dielectric slab of thickness 4 mm and dielectric constant 5 is introduced between the plates, how will
a) its capacitance
b) the electric field between the plates,
c) the energy density of the capacitor get affected?
Justify your answer in each case.